W-F-95E
September 8, 1976
SUPERSEDING
Int. Fed. Spec. W-F-0095D(GSA-FSS)
July 29, 1970
Fed. Spec. W-F-95C
January 24, 1966

FEDERAL SPECIFICATION

FAN, CIRCULATING, RADIAL DISCHARGE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers radial discharge fans (hassock fans) designed for circulation of air at floor level.

1.2 Classification.

1.2.1 Type, grade, and size. Radial discharge fans covered by this specification shall be of one type, one grade, and the following size:

Size 12 - 12 inch blade assembly.

o

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

J-C-175 - Cable Assembly, Power Electrical (3-Wire Grounding Type, For Use On 125 Volt Equipment).
QQ-Z-325 - Zinc Coating, Electrodeposited, Requirements For.
CC-M-1807 - Motors Alternating Current Fractional And Integral Horsepower (500 HP And Smaller).

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment.
Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer, and Related Materials: Methods of Inspection, Sampling, and Testing.
Fed. Test Method Std. No. 151 - Metals, Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:22Z
Check the source to verify that this is the current version before use.

Military Specification:

MIL-E-16298 - Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging Of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-461 - Electromagnetic Interference Characteristics, Requirements for Equipment.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-463 - Definitions and System of Units, Electromagentic Interference Technology.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Underwriters' Laboratories, Inc., UL Standards:

No. 507 - Standard for Electric Fans.
No. 746 - Document "E" Guide to Requirements for Polymeric Materials Used as Electrical Insulation.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, Illinois 60611; 1285 Walt Whitman Road, Melville, L.I., New York 11746; or 1655 Scott Blvd., Santa Clara, California 95050.)

National Electrical Manufacturers Association (NEMA) Publication:

Pub. No. MG1 - Motors and Generators.

(Application for copies should be addressed to National Electrical Manufacturers Association, 155 East 44th Street, New York, N.Y. 10017.)

American Society for Testing and Materials (ASTM) Standards:

B499 - Measurement of Coating Thicknesses by the Magnetic Method: Nonmagnetic Coatings on Magnetic Basis Metals.
D1730-67 - Preparation of Aluminum and Aluminum-Alloy Surfaces for Painting.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

American National Standards Institute, Inc., (ANSI) Standard:

ANSI S1.4 - 1971 - General Purpose Sound Level Meters.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, N.Y. 10018.)

3. REQUIREMENTS

3.1 Preproduction samples.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:22Z
Check the source to verify that this is the current version before use.

3.1 Preproduction sample. Unless otherwise specified (see 6.1), before production is commenced, a complete fan assembly covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2.2.1, 4.2.2.2, 4.2.3 and 4.3. Approval of the preproduction sample authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.2 Fire and casualty hazards. Each contractor shall submit to the contracting agency proof that the fans he proposes to supply under this specification conform to the requirements of UL Standard No. 507. Listing by Underwriters' Laboratories will be accepted as evidence that the product conforms to this requirement. In lieu of Underwriters' Listing, the contractor may submit independent proof satisfactory to the contracting agency that the fans conform to applicable requirements of the published standard, including method of tests of UL Standard No. 507. Compliance with the above preliminary requirements in regard to fire, safety and casualty hazards does not absolve the contractor from complete compliance with the other requirements of this specification in order to secure acceptance of his material or equipment.

3.3 Material. All material used in the manufacture of fan component parts shall be new. Additionally, material shall be free of defects that affect the fan's appearance, serviceability, or safety.

3.3.1 Plastic. Each type of plastic used in the fans construction shall be identified by the supplier. Plastic components other than slow burning or self extinguishing shall not be used in the fan's construction. When plastic is used as an enclosure for live electrical parts or to support same, it shall be of a self extinguishing type. Testing to determine plastic type shall be in accordance with 4.3.8.

3.4 Component parts.

3.4.1 General. The fan shall consist of a blade assembly connected directly to and driven by an electric motor, a protective housing for enclosing all moving parts, a means to direct airflow outward in a horizontal plane around periphery of fan housing, a speed control device, an electric cord, and all other parts necessary to constitute a completely assembled and operational radial discharge fan.

3.4.2 Motor. Unless otherwise specified (see 6.1), the fan motor shall be of a single phase, induction, squirrel cage, guarded machine design of the permanent-split capacitor, two value capacitor, or shaded pole type meeting the requirements specified in Federal Specification CC-M-1807 for type I, Class 1, Style A, Kind 2 (Group C or D) or Kind 3, Enclosure 1, Design D motors. Motor shall have torque sufficient to start and operate fan at any specified speed control setting. Additionally, motor shall be capable of operating continuously under maximum load without exceeding the temperature rise specified in UL Standard No. 507. See 4.3.3.

3.4.2.1 Voltage and frequency. Unless otherwise specified (see 6.1), motors shall be designed for a rated voltage and frequency of 115 volts, 60 HZ. or 120 volts, 60 HZ. alternating current and shall be capable of operating within the range of 115 volts ± 10 percent, 60 HZ. ± 5 percent; or 120 volts ± 10 percent, 60 HZ. ± 5 percent.

3.4.2.2 Insulation. Insulation shall be class A or B in accordance with NEMA Publication No. MG-1.

3.4.2.3 Bearings. Bearings shall be ball, sleeve, or spherical design and shall be of the maintenance free permanent lubrication type.

3.4.2.4 Duty. Motor shall be designed for continuous operation when employed as the drive source for a radial discharge fan as described herein.

3.4.2.5 Electromagnetic interference suppression. When specified in the invitation for bids (see 6.1), motors shall be designed for suppression of electromagnetic interference in accordance with requirements of MIL-STD-461, Notice 4. Testing shall be in accordance with 4.3.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:22Z
Check the source to verify that this is the current version before use.

3.4.3 Protective housing. The protective housing shall completely enclose all moving fan parts. Completely enclosed, as used herein, shall be understood to include the bottom, top, and sides of the fan. All openings in the protective housing shall be of a size that precludes entrance of a 1/2 inch diameter sphere when sphere is thrust into any opening(s) with a force of one pound (see 4.3.5.1). The overall outside dimension of the housing shall not exceed fan blade assembly diameter by more than 6 inches. The overall height of the protective housing shall be no greater than 22 inches. If protective housing is constructed of steel wire, wire shall be no smaller in diameter than AWG No. 8 (.128 inch). The supporting base or legs for securing the motor mounting bracket and protective housing shall be metal. The bottom of supporting base or legs shall be provided with a suitable material to prevent marring floor surfaces. The fan shall have sufficient strength to withstand the sandbag impact test of 4.3.5.1 without evidence of distortion. The protective housing shall be designed to preclude vibration noise, particularly vibration noise caused by any portion of the protective housing coming in contact with any other portion of housing or fan.

3.4.4 Blade assembly. The blade assembly shall consist of a complete air-screw or propeller, comprising the blades together with a hub or other central support. The blade assembly shall be balanced to an accuracy of 3 gm - cm. The blade assembly shall not crack, chip, or break when operated for one hour at a speed of 150 percent of the maximum design speed (see 4.3.6.1). Additionally, fan shall be designed so that blade assembly will not make contact with the protective housing or motor should blade assembly become loose on the motor shaft. The diameter of the blade assembly shall be 12 inches plus or minus 1/2 inch.

3.4.4.1 Fan blades. Fan blades shall be constructed of metal or plastic. Non integral metal blades shall be secured to the hub or central support with not less than two rivets per blade or other means of equal strength and rigidity. Other fan blade designs if not integral cast parts of blade assembly shall be secured to hub or central support in a manner that provides strength and rigidity equivalent to that specified above for riveted metal blades. Blades shall show no displacement with respect to hub assembly upon application of 20 pound-feet of torque as specified in 4.3.5.2.

3.4.4.2 Hub or central support. The hub or central support shall be readily replaceable on the motor shaft, forming a snug fit on the motor shaft and secured in a manner that precludes turning. Additionally, the hub or central support shall show no distortion or fracture and shall not separate from fan blades when subjected to an axial pull force of 25 pounds as described in 4.3.5.2.

3.4.5 Speed regulating device.

3.4.5.1 The speed regulating device shall have not less than 2 speed positions and these positions as well as start and stop (on and off) positions shall be identified.

3.4.5.2 The speed regulating device shall be such that low speed setting provides a fan speed of 60 to 80 percent of maximum fan speed when operating at rated voltage and frequency. See 4.3.6.1.1.

3.4.5.3 No damage to fan or speed regulating device shall occur if speed regulating device is left continuously in any speed setting.

3.4.5.4 The speed regulating device shall be nonflammable and designed to permit continuous fan operation at any speed setting.

3.4.5.5 The fan shall be capable of starting at any speed setting at a voltage 10 percent lower than rated voltage.

3.4.5.6 The speed regulating control shall be conveniently located and shall be firmly and safely secured to fan.

3.4.6 Electric cord and plug. Unless otherwise specified (see 6.1), the cable assembly shall be of any class and grade specified in Fed. Spec. J-C-175. The cable assembly shall be at least 10 feet from the plug connector to the point of entrance into the fan. If cable assembly is to be of a specific color, such color shall be specified (see 6.1), otherwise the color shall be compatible with the color of the fan.

3.5 Performance. Unless otherwise indicated, the fan performance shall be determined at the rated voltage and frequency specified (see 3.4.2.1), at normal operating temperatures of the motor in the highspeed position. See 4.3.6 for performance tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:22Z
Check the source to verify that this is the current version before use.

3.5.1 Noise. The noise level of the fan shall not exceed a maximum A weighted noise level of 60 decibels (see 4.3.6.4).

3.5.2 Input. The power input (see 4.3.6.3) at rated voltage and frequency shall be of a value that provides a thrust output per power input ratio of at least 0.0043 lbs/watt. This ratio shall be obtained by dividing measured thrust (in pounds) by the measured fan motor power input (in watts) required to produce the measured thrust.

3.5.3 Output. The output of the fan (pounds thrust) shall be not less than 0.60 pound (see 4.3.6.2).

3.5.4 Air velocity. The maximum air velocity shall be at least 500 feet per minute when tested in accordance with section 4.3.6.5 of this specification.

3.6 Finish. Unless otherwise specified (see 6.1), the finish shall be the manufacturer's standard practice provided that the color of the fan is not predominately black or gray and provided that the finish is suitable for business and home use. When specified (see 6.1), finish shall be as follows:

3.6.1 Aluminum alloy parts. Aluminum parts shall be protected for corrosion resistance by either anodizing, or covering with enamel, or lacquer coating.

3.6.2 Ferrous parts. Ferrous parts shall be protected for corrosion resistance by either chromium or zinc plating or covering with enamel or lacquer coating. No corrosion shall appear, except as noted, when tested in accordance with 4.3.7.3.

3.6.3 Coatings.

3.6.3.1 Enamel and lacquer coatings. Prior to coating, ferrous parts shall be treated in accordance with the manufacturer's corrosion inhibitive methods. Aluminum parts shall be pretreated in accordance with one or more of the methods outlined in ASTM D1730. The dry film thickness shall be between .7 and 1 mil. and shall have a dry film contrast ratio of not less than 0.98. Coatings when applied at the above thickness shall ribbon or curl without flaking or otherwise loosening from the primer or base metal. Coating flexibility shall be such that no cracking or crazing occurs when tested in accordance with 4.3.7.1.

3.6.3.2 Chromium or Zinc plating. Chromium plated finishes shall have a polished appearance. Minimum thickness of the total plating shall be 0.0004 (0.4 mils) when tested in accordance with 4.3.7.2. Zinc plated finishes shall have a polished appearance and shall be type I, class 2 in accordance with QQ-Z-325 (see 4.3.7.2).

3.7 Marking for identification.

3.7.1 The fans shall be marked with contract number and date of manufacture.

3.8 Workmanship and design. The finished fans shall be clean and conform to the quality and grade of the product established by this specification. All burrs and rough and sharp edges shall be removed. Exposed welds or seams shall be ground smooth. There shall be no flaking, peeling, or chipping of paint or any exposure of bare metal. All bolt and rivet holes shall be accurately and cleanly punched or drilled and all rivet heads neatly finished. All exposed hardware shall be capped. In general, fan shall be free from any workmanship or design defects that will affect its appearance and serviceability and any workmanship or design defects that may cause injury to the user.

3.9 Manuals. For military procurements, the contractor shall furnish with each fan shipped one copy of the manufacturer's illustrated repair manual. This manual shall include care and use instructions and a complete listing of all repair parts, the contractor's part number, the part number assigned by the actual manufacturer, name of the actual manufacturer, description of parts, prices (prevailing at time of award) and quantity of each part required. Each specific repair part shall be given manufacturer's part number regardless of model number or end item application, unless parts are interchangeable, which, in this case, manufacturer will assign identical part numbers to identical parts. Figure, drawings item plate, castings, etc., numbers may be used for cross reference only and will not be used in lieu of part numbers.

Source: https://assist.dla.mil -- Downloaded: 202640-26T11:22Z
Check the source to verify that this is the currentversion before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.1), the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certification. A certificate of compliance shall be acceptable as proof that the product being offered meets the requirements of paragraphs 3.3.1, 3.4.2 and 3.4.2.5 provided the contractor furnishes actual test results, acceptable to the Government, indicating that tests have been performed to substantiate the certification. The certificate shall state that the tests described in paragraphs 4.3.3, 4.3.4 and 4.3.8 have been performed on products manufactured from the same material and manufacturing process of the items being offered and that any proposed changes in material or process will be promptly reported to the Government. The Government reserves the right to require additional testing and certification by the contractor when such changes are made or when otherwise deemed necessary.

4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be made in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.2.1 Inspection of components and materials. In accordance with 4.1, the contractor is responsible for insuring that components and materials used were manufactured, tested, and inspected in accordance with the specified requirements of referenced subsidiary specifications and standards, or, if none, in accordance with this specification.

4.2.2 End product inspection. The sample unit for these inspections shall be one completely fabricated fan.

4.2.2.1 Visual examination. The completely assembled end item shall be examined for defects in finish, construction, workmanship, and marking. Defects shall be classified as outlined below. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

Examine	Defect	Major	Minor
Finish	Not finished as required	X	
Design	Any characteristics not in accordance with the specified requirements	X	
Construction and workmanship general (applicable to all components & assemblies)	Fractured, split, bowed, malformed, or otherwise impaired	X	
	Sharp burrs, slivers, or splinters that may be injurious to personnel	X	
Assembly	Not properly assembled or secured		X
Marking identification	Missing, incomplete, not legible, not as specified		X

4.2.2.2 Dimensional examination. Examination shall be made of the end item to determine compliance with the dimensional requirements. Any noncompliance with a specified requirement shall constitute a defect. The inspection level shall be S-2 and the AQL shall be 6.5, expressed in terms of defects per hundred units.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:22Z
Check the source to verify that this is the currentversion before use.

4.2.2.3 Testing of the end product. The contractor shall perform tests in accordance with 4.3.2 using inspection level S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.2.3 Code and standards compliance. The contractor shall provide the Government representative with proof of compliance with the requirements of 3.2.

4.3 Tests.

4.3.1 Preproduction sample tests. Preproduction sample tests shall consist of all tests specified or described under 4.3. Evidence acceptable to the Government and indicating specification compliance may be offered in lieu of fan manufacturer performance of 4.3.3, 4.3.4 and 4.3.8 tests. Except as noted for 4.3.3, 4.3.4 and 4.3.8, all tests are to be conducted on the same fan sample at manufacturer's expense.

4.3.2 Lot acceptance tests. Unless otherwise indicated the contractor shall furnish all test samples and shall be responsible for accomplishing the required tests. Testing, if conducted at the contractor's plant, shall be under the supervision of the Government inspector. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a testing laboratory acceptable to the Government. Except as indicated, each fan selected for lot acceptance testing shall be subjected to all tests cited below. Lot acceptance tests shall consist of all tests cited in 4.3.3, 4.3.4 (when required), 4.3.5, 4.3.6, 4.3.7 (when required), 4.3.8 and 4.3.9; however evidence acceptable to Government and indicating specification compliance may be offered in lieu of fan manufacturer performance of 4.3.3, 4.3.4 and 4.3.8 tests.

4.3.3 Motor. With exception of temperature rise test, motor tests shall be conducted in accordance with requirements of Federal Specification CC-M-1807. Temperature rise test shall be in accordance with UL Standard No. 507. Failure of above tests shall be cause for rejection of product. Refer to 3.4.2 for motor requirements.

4.3.4 Electromagnetic interference test. When specified (see 3.4.2.5 and 6.1), electromagnetic interference suppression tests shall be conducted in accordance with MIL-STD-461, Notice 4 and MIL-STD-462. Refer to MIL-STD-463, as necessary, for definitions and systems of units used in electromagnetic interference technology. The test sample shall be one complete fan. Testing shall be accomplished by a test laboratory acceptable to the Government. Failure shall be cause for rejection of product. Refer to 3.4.2.5 for interference suppression requirements.

4.3.5 Structural tests.

4.3.5.1 Protective housing. To determine compliance with requirements of 3.4.3 the following tests shall be performed. A 1/2 inch diameter sphere shall be thrust into any opening(s) of fan housing with a force of one pound from a direction and in a manner most apt to favor penetration of the protective housing. Sphere penetration of protective housing shall be cause for test failure. Subsequent to sphere penetration test, the completely assembled fan shall be set on a rigid horizontal surface, whereupon a sandbag having a diameter approximately the same as that of the fan top and weighing 150 pounds shall be dropped in free fall from a height of six inches onto the top of the fan. The impact shall be repeated 6 times, following which, the frame, legs, and connecting members shall show no evidence of distortion. Evidence of distortion shall be cause for test failure.

4.3.5.2 Blade torque and hub axial pull force tests. To determine compliance with the blade torque requirements of 3.4.4.1, each blade of the blade assembly shall be subjected to a torque of 20 pound-feet applied in such manner that the force causing the torque is directed in the plane of blade rotation. Torque shall be applied for one minute whereupon blades shall be examined for displacement from the hub assembly. To determine compliance with the axial pull force requirements of 3.4.4.2, the blades of the complete blade assembly shall be held in a fixed position whereupon an axial pull force of 25 pounds shall be applied to the hub or central support for one minute alternately in both axial directions. After force application the hub or central support shall be examined for evidence of displacement distortion or fracture. Evidence of displacement, distortion or fracture after performance of either of the foregoing tests shall be construed as test failure and shall be cause for rejection.

4.3.6 Performance tests. Except as noted, tests as specified or described under 4.3.6 shall be conducted at rated voltage and frequency after the fan has been in operation for not less than one hour.

Source: https://assist.dla.mil -- Downloaded: 2026l0-26T11:22Z
Check the source to verify that this is the currentversion before use.

4.3.6.1 Overspeed. The blade assembly shall be removed from the fan motor-shaft and secured to a variable speed motor having a high-speed rating of at least 175 percent of the fan high-speed rating. The blade assembly shall be run at a speed of 150 ± 3 percent of the fan high-speed rating for a period of one hour. Upon completion of the test, the blade assembly shall be examined for cracks, breaks, chipping, and tightness of blades to the hub or central support. The speed of rotation shall be measured stroboscopically or with a counter that imposes no appreciable load on the motor. Cracks, breaks, chipping or signs of distortion shall be construed as test failure and shall be cause for rejection. Refer to 3.4.4 for requirements.

4.3.6.1.1 Speed. To determine compliance with 3.4.5.2, the fan shall be operated at both low and maximum speed settings and speed (r.p.m.) shall be measured stroboscopically or with a counter which imposes no appreciable load on fan motor. The sensitivity and accuracy of the test apparatus shall be such that difference of a 20 r.p.m. may be detected. Fan may be partially disassembled as appropriate for test performance. Inability to meet requirements of 3.4.5.2 shall be construed as test failure and shall be cause for rejection.

4.3.6.2 Measurement of thrust. Refer to 3.5.3 for thrust requirements. Inability to meet thrust requirements shall be construed as test failure and shall be cause for rejection.

4.3.6.2.1 Procedure. The assembled motor, motor mounting bracket, and blade assembly shall be removed from the protective housing and mounted on the thrust measuring apparatus. The fan shall be mounted in the testing apparatus in a manner that will place the motor shaft in the horizontal position. Thrust shall then be determined at maximum speed setting.

4.3.6.2.2 Sensitivity of apparatus. The thrust shall be measured by means of an apparatus having a sensitivity such that a change in thrust of 0.01 pound causes a motion of the pointer or indicating device of at least 1/8 inch.

4.3.6.2.3 Description of apparatus. The apparatus shall be calibrated by means of a standard weight. In case of controversy, the apparatus developed at the National Bureau of Standards shall be used. This apparatus is based on the principle of a pendulum, counterbalanced to give adequate sensitivity. A platform is suspended by four rods of equal length, the connection of the rods to the platform and to the supporting framework being through ball bearings. The weight of the platform and a part of the weight of any fan placed on the platform are counterbalanced by two heavy counterweights placed above the plane of the four upper supporting bearings. In practice, the overhead weights are adjusted to give an overbalance condition with no fan; and the desired stability and sensitivity are obtained by adding weights to the platform. The current for the fan motor is brought in through the frame and through an insulated line carried through insulated brush contacts. A scale graduated in terms of the tangent of the deflection angle is attached to the platform.

4.3.6.2.4 Calibration of apparatus. The scale is calibrated and the sensitivity measured by applying horizontal forces by means of a light thread running over a ball-bearing pulley to standard weights.

4.3.6.2.5 Alternate method of determing thrust. In lieu of determing thrust by the above method, the following alternate method may be used. Using the apparatus described in 4.3.6.2.3, with exception of graduated scale pointer device, place the fan on the platform with the motor axis in a horizontal plane and mark the platform's initial position with a stationary marker. After marking position of platform, attach a light thread to the bottom of the platform. Run thread horizontally to a ball bearing pulley. Run thread over pulley and downward to a position below the platform. At this point, attach a bucket, containing lead weights, to the thread. Turn fan on and set to highest speed position. Change weight in bucket until platform is returned to the initial position as marked above. Detach bucket and weigh on a scale or balance having an accuracy of at least 0.001 pounds. The weight of the bucket and lead weights so obtained gives the direct measurement of fan thrust.

4.3.6.3 Power input. The sensitivity and accuracy of the apparatus for measuring power input to determine compliance with power input requirements of 3.5.2 shall be such that differences of one watt may be determined. Power input shall be determined at maximum speed. Inability to meet power requirements shall be construed as test failure and shall be cause for rejection.

4.3.6.4 Noise test. Refer to 3.5.1 for noise requirement. Test shall be accomplished at all speed settings. Inability to meet noise requirement shall be construed as test failure and shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:22Z
Check the source to verify that this is the current version before use.

4.3.6.4.1 Noise measurement. Noise measurement shall be made with a general purpose type 2 sound level meter conforming to ANSI Standard S 1.4-1971 or latest revision thereof. The sound level meter shall conform to response curve A of the above ANSI S 1.4 Standard.

4.3.6.4.2 The fan shall be mounted in its normal operating position centrally in the test room.

4.3.6.4.3 Resilient means shall be interposed between fan and test room floor.

4.3.6.4.4 Five measurements shall be taken at stations in one horizontal plane which shall be four feet above the room floor. These stations shall all be spaced equidistant on a circle which is concentric with the vertical axis of rotation of the fan shaft and is four feet in diameter.

4.3.6.4.5 The fan speed at the beginning of the noise test shall be kept constant during the test.

4.3.6.4.6 The average of the readings shall be the noise level of the fan at the test speed. If the range of the sound level measurement is less than 10 decibels, their average can be obtained by an arithmetical average of the decibel readings. However, the average of a number of sound level readings is defined as 10 times the logarithm to the base 10 of the average of the antilogarithms of 1/10 of the separate decibel values; this value to be used where differences in reading between positions exceed 10 decibels. The following example will illustrate the method of obtaining the average:

Microphone	Sound	Antilogarithm to the base 10 of 1/10 of the decibel sound level
	Decibels	
A	60	1,000,000
B	50	100,000
C	40	10,000
	Total.....	1,110,000
	Average.....	370,000

Log10 370,000 = 5.57.
10 Log10 370,000 = 55.7 or the average decibel value.

4.3.6.4.7 If the noise level of the room when the fan is not in operation is less than 10 decibels lower than any reading taken with the fan in operation, the reading shall be corrected in accordance with the following:

Difference in decibels between room level and fan level reading	Subtract this decibel correction from fan level reading
6	1.3
7	1.0
8	0.8
9	0.6
10	0.4

4.3.6.4.8 Test room. The test room for measuring the noise shall be so constructed that the noise reflected from the walls, floor, and ceiling will have a negligible effect on the measurements taken. The size of the test room shall be such that recirculating air currents will not affect the motor speed or the power input. The noise level of the test room when the fan is not in operation, shall be at least 6 decibels lower than the lowest reading which is obtained during the operation of the fan.

4.3.6.5 Air velocity. The fan shall be mounted on a turntable platform, the surface of which shall be 18 inches above the test floor. The turntable shall be located in the center of a test space having a minimum ceiling height of 8 feet and a clear space of not less than 20 feet between walls. The center of rotation of the test fan and turntable shall coincide. The plane of rotation of the anemometer wheel shall be 18 inches from the fan axis. The turntable platform shall be circular and its diameter shall be at least 1 foot larger than the greatest overall horizontal dimension of the fan. Refer to 3.5.4 for requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:22Z
Check the source to verify that this is the current version before use.

4.3.6.5.1 Air velocities shall be measured with an anemometer that has been calibrated by the National Bureau of Standards or an accredited commercial testing laboratory. Anemometer readings shall be taken in 4 different vertical planes. The 4 planes shall be equally spaced over the circumference of the fan. Anemometer readings shall be taken at stations located in each of the 4 vertical planes. Each station reading shall be taken over an operating period of not less than 2 minutes. All readings shall be taken at maximum fan speeds.

4.3.7 Finish tests. Inability to meet finish requirements when tested as specified below shall be construed as test failure and shall be cause for rejection.

4.3.7.1 Enamel and lacquer. Dry film thickness and contrast ratio shall be determined in accordance with test methods 6183 and 4122.1 respectively of Fed. Test Method Standard 141. Ability to ribbon or curl without flaking or loosening shall be determined in accordance with Method 6204.1 of Fed. Test Method Standard No. 141. Compliance with the flexibility requirement of 3.6.3.1 shall be determined by rapidly doubling a coated tinplate test panel over a 1/8-inch diameter steel rod mandrel in accordance with Method 6221 of Fed. Test Method Standard 141. Prior to doubling the coated panel shall be: (1) allowed to air dry for 2 hours at a temperature of 23° ± 1.1° C. (73.4° ± 2°F.) and a relative humidity of 50 ± 4 percent; (2) baked for 3 hours at 105°C.; and (3) conditioned for at least 1/2 hour at a temperature of 23° ± 1.1°C. (73.4° ± 2°F.) and a relative humidity of 50 ± 4 percent. Refer to 3.6.3.1 for enamel and lacquer coating requirements.

4.3.7.2 Thickness of Chromium and Zinc finishes. Determine total thickness of chromium plated surfaces and zinc plated surfaces by either the magnetic test of ASTM B499 or the electronic test, Method 520.1 of Fed. Test Method Std. No. 151 (see 3.6.3.2).

4.3.7.3 Resistance to rust test (ferrous parts). All grease is removed from the parts to be tested, by immersion in 1,1,1-trichlorethane for 10 minutes. The parts are then immersed for 10 minutes in a 10 percent solution of ammonium chloride in water at a temperature of 20 ± 5°C. Without drying, but after shaking off any drops, the parts are placed for 10 minutes in a box containing air saturated with moisture at a temperature of 20 ± 5°C. After the parts have been dried for 10 minutes in a heating cabinet at a temperature of 100 ± 5°C, their surfaces shall show no signs of rust. Traces of rust on sharp edges and any yellowish film removable by rubbing are ignored. See 3.6.2.

4.3.8 Burning rate/self extinguishing tests (plastic parts). Testing to determine plastic type, either slow burning or self extinguishing, shall be in accordance with Underwriter's Laboratories Inc., "Burning Test For Polymeric Materials Classed As Slow-Burning", and "Burning Test For Polymeric Materials Classed As Self-Extinguishing", reference pages E3, E4, and E5 of UL subject 746 of March 1, 1967. Refer to 3.3.1 for requirements. Failure to meet requirements of paragraph 3.3.1, as determined per this test shall be cause for rejection of product.

4.3.9 Inspection of preparation for delivery. The preservation, packaging, packing, and marking of the fans shall be examined and tested to determine compliance with MIL-E-16298 and section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Each fan shall be preserved, packaged, and packed in accordance with MIL-E-16298 for levels as specified in contract or order (see 6.1).

5.2 Marking.

5.2.1 Civil agencies. In addition to special marking required by contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. Packages and shipping containers shall be marked in accordance with MIL-STD-129.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:22Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) If preproduction sample not required (see 3.1.2).
(c) Motor characteristics, if other than specified (see 3.4.2).
(d) Operating voltage and frequency if other than specified (see 3.4.2.1).
(e) If supression of electromagnetic interference is required (see 3.4.2.5).
(f) Cable assembly, if other than specified and color of cable assembly, if other than specified (see 3.4.6).
(g) Finish, if other than manufacturer's standard practice or if the specific finish indicated in 3.6 is desired (see 3.6).
(h) Inspection responsibility, if other than specified (see 4.1).
(i) Levels of preservation, packaging, and packing (see 5.1).
(j) Special marking, if required (see 5.2).

6.2 Additional details for the thrust measurement apparatus discussed in paragraph 4.3.6.2, with description and working drawings, may be obtained from the National Bureau of Standards, Washington, D.C.

MILITARY INTERESTS:

Custodians:

Army - ME
Air Force - 82

Review Interest:

Defense General Supply - GS

Coordinating Activity:

Army - ME

Preparing Activity:

GSA-FSS

CIVIL AGENCY COORDINATING ACTIVITIES:

USDA - HEW
VA -

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 35 cents each.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:22Z
Check the source to verify that this is the current version before use.